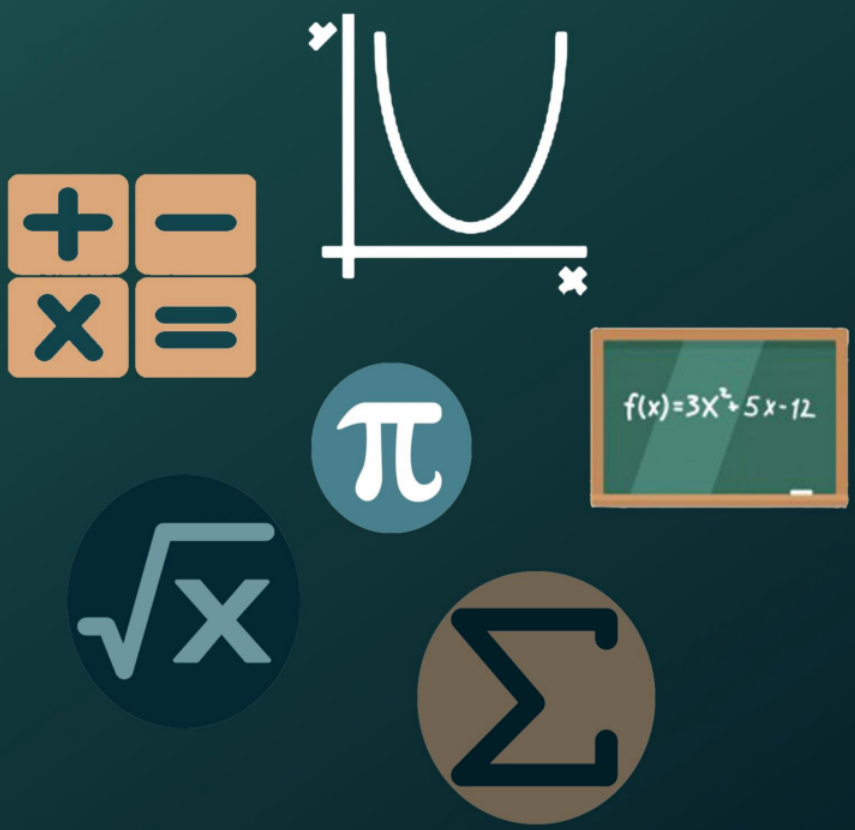
MATHEMATICS

Relation



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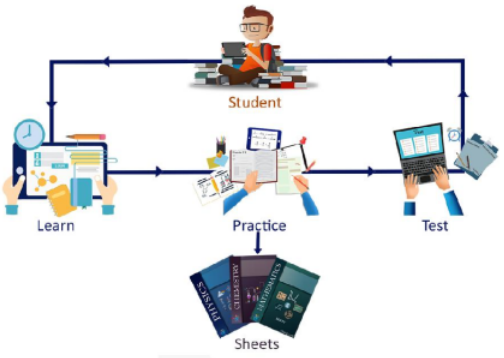
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-2
1	Introduction to Relations	1, 2	1-4
2	Types of Relations		5
3	Classification of Relations	3-8	6-13
4	After Complete Chapter	9-21	

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Relations

Q1. If A = {2, 3} and B = {1, 2}, then A × B is equal to-

(A) {(2, 1), (2, 2), (3, 1), (3, 2)}

(B) {(1, 2), (1, 3), (2, 2), (2, 3)}

(C) {(2, 1), (3, 2)}

(D) {(1, 2), (2, 3)}

Q2. Let R be a relation from a set A to a set B, then-

(A) $R = A \cup B$ (B) $R = A \cap B$

(C) $R \subseteq A \times B$ (D) $R \subseteq B \times A$

Classification of Relations

Q3. For real numbers x and y, we write $x\ R\ y \Leftrightarrow x - y + \sqrt{2}$ is an irrational number. Then the relation R is-

(A) Reflexive (B) Symmetric

(C) Transitive (D) None of these

Q4. Let L denote the set of all straight lines in a plane. Let a relation R be defined by $\alpha\ R\ \beta \Leftrightarrow \alpha \perp \beta$, $\alpha, \beta \in L$. Then R is-

(A) Reflexive (B) Symmetric

(C) Transitive (D) None of these

Q5. Which one of the following relations on R is equivalence relation-

(A) $x\ R_1 y \Leftrightarrow |x| = |y|$

(B) $x\ R_2 y \Leftrightarrow x > y$

(C) $x\ R_3 y \Leftrightarrow x / y$ is integer

(D) $x\ R_4 y \Leftrightarrow x < y$

Q6. Let $x, y \in I$ and suppose that a relation R on I is defined by x R y if and only if $x \leq y$ then

(A) R is partial order ralation

(B) R is an equivalence relation

(C) R is reflexive and symmetric

(D) R is symmetric and transitive

Q7. Given the relation R = {(1, 2), (2, 3)} on the set A = {1, 2, 3}, the minimum number of ordered pairs which when added to R make it an equivalence relation is -

(A) 5 (B) 6

(C) 7 (D) 8

Q8. Let A = {p, q, r}. Which of the following is an equivalence relation in A ?

(A) R_1 = {(p, q), (q, r), (p, r), (p, p)}

(B) R_2 = {(r, q) (r, p), (r, r), (q, q)}

(C) R_3 = {(p, p), (q, q), (r, r), (p, q)}

(D) None of these

Q9. A={1,2,3,5}, B={4,6,9}. Define a relation R from A to B by R = {(x,y); the difference between x and y is odd; $x \in A$, $y \in B$}.

Then R is :

(A) {(1,2),(1,6),(2,9),(3,4),(3,5)}

(B) {(1,4),(1,6),(2,9),(3,4),(3,6),(5,4),(5,6)}

(C) {(1,6),(1,7),(1,8),(2,9),(2,4),(2,9)}

(D) {(1,5),(1,6),(1,7),(6,4),(6,9),(6,2)}

Q10. Given A = {1,2,3,4,5}, S = {(x,y) : $x \in A$, $y \in A$}. Then the ordered pair which satisfy $x + y > 8$ then R is .....

(A) {(5,5) (5,3)}

(B) {(4,5) (5,4) (5,5)}

(C) {(5,4) (5,6)}

(D) {(5,4) (5,3)}

Q11. The domain and range of

$$R = \{(x, y) / y = x + \frac{6}{x}, x, y \in N \text{ and } x < 6\}.$$

(A) Domain = {1, 2}, Range = {7, 5}

(B) Domain = {1, 2, 3}, Range = {7 , 5}

(C) Domain = {1}, Range = {7, 5}

(D) Domain = {1, 2}, Range = {7}

Q12. If the number of relations on a finite set A having 'n' elements is 2^{16}, then 'n' equal to:

(A) 15 (B) 17

(C) 4 (D) 8

Q13. On the set of natural numbers N, the relation R is defined by xRy iff x + y = 100 is -

(A) reflexive (B) not reflexive

(C) equivalence (D) not symmetric

Q14. On the set of all vectors in space the relation R is defined by $\vec{a} R \vec{b} \Leftrightarrow \vec{a}.\vec{b}$ is scalar is -

(A) symmetric (B) not symmetric

(C) not reflexive (D) both B and C

Q15. If A={ 1, 2,3} Then a relation reflexive but not Symmetric on A is -

(A) {(1,1), (1,2)}

(B) {(1,1),(1,2),(2,1),(2,2)}

(C) {(1,1),(2,2),(3,3)}

(D) {(1,1),(2,2),(3,3),(2,3)}

Q16. Let A be the set of first 10 natural numbers and let

$$R = \{(x, y) / x \in A, y \in N \text{ and } x + 2y = 10\}$$

then $n\{dom(R^{-1})\} =$

(A) 4 (B) 5

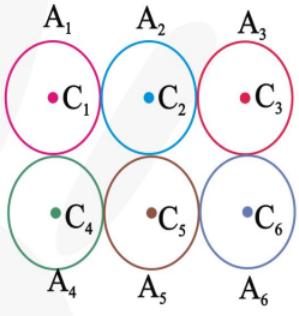
(C) 8 (D) 10

Q17. Let $X = \{1, 2, 3, 4, 5\}$, the number of different ordered pairs (Y,Z) that can be formed such that $Y \subseteq X$, $Z \subseteq X$ and $Y \cap Z$ is empty, is -

(A) 3^5 (B) 2^5

(C) 5^3 (D) 5^2

Q18. Let $A = \{A_1, A_2, A_3, A_4, A_5, A_6\}$ be the set of six unit circles with centres $C_1, C_2, C_3 ... C_6$ arranged as shown in the diagram.The relation R on A is defined by $(A_i, A_j) \in R \Leftrightarrow C_i C_j \leq 2\sqrt{2}$ then -



(A) R is symmetric, transitive but not reflexive

(B) R is only transitive

(C) R is symmetric, reflexive but not transitive

(D) R is neither reflexive nor transitive but is symmetric.

Q19. The relation R on the set of natural numbers N is defined as $xRy \Leftrightarrow x^2 - 4xy + 3y^2 = 0$, $x, y \in N$ then R is -

(A) reflexive but not symmetric and not transitive

(B) symmetric but not reflexive and not transitive

(C) transitive but not reflexive and not symmetric

(D) equivalence relation

Q20. A relation R on the set of non zero complex numbers is defined by $z_1 R z_2 \Leftrightarrow \frac{z_1 - z_2}{z_1 + z_2}$ is real, then R is -

(A) Reflexive (B) Symmetric

(C) Transitive (D) Equivalence

Q21. S is a relation over the set R of all real numbers and it is given by (a, b) $\in$ S $\Leftrightarrow$ ab $\geq$ 0. Then S is -

(A) symmetric and transitive only

(B) reflexive and symmetric only

(C) a partial relation

(D) an equivalance relation

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If R is a relation from a finite set A having m elements to a finite set B having n elements, then the number of relations from A to B is-

(A) 2^{mn} (B) $2^{mn}-1$

(C) 2mn (D) m^n

Q2. Let X = {1, 2, 3, 4} and Y = {1, 3, 5, 7, 9}. Which of the following is relations from X to Y-

(A) $R_1 = \{(x, y) \mid y = 2 + x, x \in X, y \in Y\}$

(B) $R_2 = \{(1, 1), (2, 1), (3, 3), (4, 3), (5, 5)\}$

(C) $R_3 = \{(1, 1), (1, 3), (3, 5), (3, 7), (5, 7)$

(D) $R_4 = \{(1, 3), (2, 5), (2, 4), (7, 9)\}$

Q3. A and B are two sets having 3 and 4 elements respectively and having 2 elements in common. The number of relations which can be defined from A to B is-

(A) 2^5 (B) $2^{10}-1$

(C) $2^{12}-1$ (D) 2^{12}

Q4. The relation R defined in A = {1, 2, 3} by a R b if $|a^2 - b^2| \leq 5$. Which of the following is false-

(A) R ={(1, 1), (2, 2), (3, 3), (2, 1), (1, 2), (2, 3), (3,2)

(B) $R^{-1} = R$

(C) Domain of R = {1, 2, 3}

(D) Range of R = {5}

Q5. If R be a relation '<' from A = {1, 2, 3, 4} to B = {1, 3, 5} i.e. (a, b) $\in$ R iff a < b, then ROR^{-1} is-

(A) {(1, 3), (1, 5), (2, 3), (2, 5), (3, 5), (4, 5)}

(B) {(3, 1), (5, 1), (3, 2), (5, 2), (5, 3), (5, 4)}

(C) {(3, 3), (3, 5), (5, 3), (5, 5)}

(D) {(3, 3), (3, 4), (4, 5)}

Q6. Let R be a relation defined in the set of real numbers by a R b $\Leftrightarrow$ 1 + ab > 0. Then R is-

(A) Equivalence relation

(B) Transitive

(C) Symmetric

(D) Anti-symmetric

Q7. For n, m $\in$ N, n|m means that n is a factor of m, the relation | is-

(A) reflexive and symmetric

(B) transitive and symmetric

(C) reflexive, transitive and symmetric

(D) reflexive, transitive and not symmetric

Q8. Let R = {(x, y) : x, y $\in$ A, x + y = 5} where A = {1, 2, 3, 4, 5} then

(A) R is not reflexive, symmetric and not transitive

(B) R is an equivalence relation

(C) R is reflexive, symmetric but not transitive

(D) R is not reflexive, not symmetric but transitive

Q9. Let R be a relation on a set A such that $R = R^{-1}$ then R must be

(A) reflexive (B) symmetric

(C) transitive (D) None of these

Q10. Let X be a family of sets and R be a relation on X defined by 'A is disjoint from B'. Then R is-

(A) reflexive

(B) symmetric

(C) anti-symmetric

(D) transitive

Q11. Let N denote the set of all natural numbers and R be the relation on N × N defined by (a, b) R (c, d) if ad (b + c) = bc(a + d), then R is-

(A) Symmetric only

(B) Reflexive only

(C) Transitive only

(D) An equivalence relation

Q12. Let W denote the words in the English dictionary.

Define the relation R by :

R = {(x, y) ∈ W × W : the words x and y have at least one letter in common}.

Then R is-

(A) reflexive, symmetric and not transitive

(B) reflexive, symmetric and transitive

(C) reflexive, not symmetric and transtive

(D) not reflexive, symmetric and transitive

Q13. Let R and S be two equivalence relations in a set A. Then-

(A) R ∪ S is an equivalence relation in A

(B) R ∩ S is an equivalence relation in A

(C) R – S is an equivalence relation in A

(D) none of these

Answer Key

Ex - 1									
1	2	3	4	5	6	7	8	9	10
A	C	A	B	A	A	C	D	B	B
11	12	13	14	15	16	17	18	19	20
B	C	B	A	D	A	A	C	A	D
21									
D									

Ex-2									
1	2	3	4	5	6	7	8	9	10
A	A	D	D	C	C	D	A	B	B
11	12	13							
D	A	B							